

Kasra Mossayebi

Mathematics undergraduate working in higher algebra, homotopy theory, Hodge theory, and quantum information

📍 Toronto, Ontario, Canada ✉ kasra.mossayebi@mail.utoronto.ca 🌐 kasramossayebi.github.io

🔗 Kasramossayebi 🆔 0009-0004-6714-3433

Education

BSc	University of Toronto , Mathematics Specialist	Toronto, Canada Sept 2025 – present
	- cGPA: 3.85 - Coursework includes analysis, algebra, topology, and advanced ordinary differential equations. - Member of the directed reading group on infinity-categories in algebraic geometry.	

Research Experience

Northeastern University / IAIFI , Algebraic K-Theory for Resources	Boston, United States June 2026 – present
- Applied Waldhausen's S-construction to quantum-resource categories to seek computable spectra and invariants. - Explored Segal and May delooping machines and applications to quantum magic. - Preprint in preparation.	
Northeastern University / IAIFI , A-infinity Algebras and Entanglement	Mar 2026 – present
- Extended the entanglement DGA to an A-infinity algebra via homotopy transfer. - Established local-unitary invariance and extracted basis-independent invariants from higher Massey products. - Preprint in preparation.	
With Christian Ferko , Extending Generative Effects to Entanglement	Dec 2024 – present
Explored categorical, operator-algebraic, and higher-descent approaches to detecting entangled states.	
With Christian Ferko, Eashan Iyer, and Gregor Sanfey , A Hodge Theory for Entanglement Cohomology	July 2024 – Jan 2025
- Developed a Hodge-theoretic framework for entanglement cohomology and proved compatibility of support operators. - Published in Physical Review A.	
Independent project , Homotopical Equivalence of Categories	Mar 2023 – July 2023
Studied conditions for adjoint equivalences of relative categories to extend to homotopical equivalences.	

Publications

A Hodge Theory for Entanglement Cohomology	Mar 2025
Christian Ferko, Eashan Iyer, Kasra Mossayebi , Gregor Sanfey	
10.1103/PhysRevA.111.032422 (Physical Review A)	

Talks

An Introduction to Entanglement Cohomology	Physics Undergraduate Colloquium, University of Toronto
Introduced the GNS entanglement complex and used a two-qubit example to show how it detects entanglement.	Jan 2026

Generative Effects and Quantum Entanglement

Presented a categorical perspective on generative effects and possible extensions to entanglement and higher-category descent.

Mathematics Undergraduate Colloquium, University of Toronto
Oct 2025

Awards

Ted Mossman Scholarship

Awarded for showing exceptional promise.

University College, University of Toronto
Oct 2025